

STUDY GUIDE

GxP Training Bot

Understand your own project — slide by slide, word by word

Read Part 1 first. It explains the whole project in plain English with no jargon at all. Everything after that will make sense once Part 1 does.

Part 2 tells you what to say on each slide. Part 3 explains every technical word. Part 4 is the questions you will be asked.

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Part 1 — The Whole Project in Plain English

Read this part once, slowly. It uses no technical words. If you understand only this page, you can still answer most questions an examiner will ask.

1.1 The real-world problem

Medicines are made in factories that are heavily regulated, because a mistake can hurt someone. Every task in such a factory is written down in a document called a Standard Operating Procedure, or SOP. An SOP might explain how to gown up before entering a clean room, or what to do when something goes wrong.

Every worker must be trained on the SOPs for their job, and re-trained regularly. Today that training is usually one set of slides and one quiz, given to everybody.

Here is the problem in one sentence:

If a worker scores 60% on the quiz, you know THAT PERSON struggled. You do not know WHAT they struggled with. So when you re-train them, you make them repeat the whole procedure again — including the parts they already knew perfectly.

That wastes the worker's time and tells the trainer nothing useful.

1.2 What your project does

Your project is a web application that fixes this in four steps.

Step	What happens	Why it matters
1. Split the document	The SOP is uploaded and automatically cut into its individual sections — for example, three sections in the demo.	Everything later depends on knowing which section a question came from.
2. AI writes questions	An AI language model reads ONE section at a time and writes quiz questions about only that section.	Each question is permanently linked to the section it came from.
3. A human expert approves	Every AI-written question goes to a qualified expert who must approve it, re-entering their password to sign it off.	The AI can make mistakes. The human is the safety check. Nothing reaches a worker unapproved.
4. The system adapts	The worker takes the quiz. Because each question is linked to a section, the system knows which SECTIONS they got wrong — and re-trains only those.	This is the actual contribution of the project.

1.3 The one thing that makes it work

Every question remembers which section of the SOP it came from.

That single link is the whole idea. Because of it, when a worker answers a question, the system does not just learn that they got one wrong — it learns that they are weak on the CAPA section specifically. Without that link, all you could ever calculate is one score for the whole document, which is exactly what existing systems already do.

1.4 The demo, told as a story

This is the story your demonstration tells. Learn this story and you can present the project.

What happens	The numbers
An SOP is uploaded. It has three sections: Good Manufacturing Practice, CAPA and Root Cause Analysis, and Documentation Practices.	3 sections
The AI writes three questions about each section.	9 questions
An expert approves all nine under an electronic signature.	9 approved
A learner takes the quiz and does badly — they get the first section right but fail the other two.	33% (3 of 9)
The system works out, section by section, where they are weak. Two sections are weak. One is strong.	2 weak, 1 strong
Re-training is built from ONLY the weak sections. The strong section is left out completely.	6 questions, not 9
The learner re-takes the targeted quiz and does well.	100%
The system measures the improvement from their stored answers.	50% → 100% (+50 points)

The line that matters most: "Six questions, not nine." That is the visible proof that the system adapted. The learner was never made to repeat what they had already mastered.

1.5 Three things you must never claim

Examiners respect accuracy far more than big claims. These three sentences protect you.

Never say	Say instead	Why
"It is GxP compliant"	"It uses GxP-oriented controls"	Real compliance needs formal validation and auditing that this project has not undergone.
"It is machine learning" / "a trained model"	"The decision layer is rule-based; two parts learn from data online"	Nothing was trained on a dataset. Claiming ML and being caught is far worse than explaining what it really is.
"It proves learning improves"	"It is a verified controlled demonstration result"	It is one learner in one controlled scenario, not a study with a control group.

Part 2 — What to Say on Each Slide

For each slide: what it shows, what to say out loud, and the technical words on that slide (all of which are explained in Part 3). Keep to roughly the time shown and the whole talk lands at about twelve minutes.

Slide 1 — Title (40 sec)

What is on the slide: The project name and a strip showing SOP → AI → SME → Learner → Adaptive Training.

What to say:

"This is GxP Training Bot. It takes a pharmaceutical Standard Operating Procedure, uses AI to write assessment questions from it, requires a human expert to approve every question, and then uses each learner's performance to decide what they should be re-trained on. The strip along the bottom is the flow of the whole system."

Words on this slide (see Part 3): SOP · SME · Adaptive training

Slide 2 — Problem Statement (60 sec)

What is on the slide: Traditional training versus this system, and the research question in a box.

What to say:

"Traditional SOP training gives everyone the same material and the same quiz. If someone scores 60%, you know who struggled but not what with. So re-training repeats everything, including what they already knew. My project asks: can SOP training be automated, while still being checked by a human, explainable, and adapted to each learner?"

Words on this slide (see Part 3): Personalisation · Traceability

Slide 3 — Aim and Objectives (50 sec)

What is on the slide: The aim in a box, then six objective cards.

What to say:

"The aim is to build an AI-assisted, GxP-oriented platform that turns SOPs into expert-reviewed assessments and then adaptively re-trains each learner's specific gaps. The first four objectives build the content pipeline. The last two are the adaptive part, which is my actual contribution."

Words on this slide (see Part 3): GxP-oriented · Chunking · Electronic signature

Slide 4 — Literature Review (70 sec)

What is on the slide: A table of seven research papers.

What to say:

"Do not read the table. Say: the literature gave me three things — a way to model ability, which is Elo; a way to model forgetting, which is FSRS; and a warning. That warning is the last row: a review of 71 studies found error rates in AI-generated questions ranging from under one percent up to forty-five percent, and concluded they should not be used without human supervision. That is exactly why expert approval is compulsory in my system."

Words on this slide (see Part 3): Elo · FSRS · Knowledge tracing · Deep Knowledge Tracing

Slide 5 — Existing vs Proposed (50 sec)

What is on the slide: Two columns, and a highlighted line at the bottom.

What to say:

"The left column is how training works today; the right is my system. But the important part is the line at the bottom: the unit of adaptation is the SOP section, not the whole document. Every question remembers which section it came from, so every answer is evidence about that specific passage."

Words on this slide (see Part 3): Unit of adaptation · Foreign key / source link

Slide 6 — System Architecture (80 sec)

What is on the slide: A diagram: top row is the content pipeline, then it splits into two engines, then loops back.

What to say:

"The top row is the content pipeline — SOP in, split into chunks, AI writes questions, then it stops at the expert review gate. Nothing passes that gate without a human. Then it reaches the learner, and here is the important split: two separate engines. One decides WHAT to train. The other decides WHEN to review it. The arrow at the bottom is what makes this a loop rather than a report."

Words on this slide (see Part 3): Django · React · Celery · NVIDIA NIM · RBAC

Slide 7 — Methodology (55 sec)

What is on the slide: Six numbered stages.

What to say:

"Six stages. Stages one to four are a content supply chain; five and six are the adaptive loop. Two points worth making: stage four is a gate, not a step — content cannot skip it. And stage five, the grading, happens entirely on the server, because if grading happened in the browser, every number after it would be untrustworthy."

Words on this slide (see Part 3): Server-side scoring · Fallback generator

Slide 8 — AI Question Generation (70 sec)

What is on the slide: A chain of steps, then two boxes — one green, one red.

What to say:

"That chain is how a question is made. But the two boxes are the point. The left one says what chunking buys me: the question comes from specific SOP text, not from whatever the model happens to know. The right one is what it does NOT buy me: provenance does not prove the answer is correct. Nothing automatically checks that the correct answer actually follows from the passage. That is what the human is for."

Words on this slide (see Part 3): Provenance · Entailment · Hallucination · Deduplication

Slide 9 — SME Review and GxP-Oriented Control (70 sec)

What is on the slide: The approval flow and a list of five controls, with a red bar at the bottom.

What to say:

"Why keep a human if the point was automation? Two reasons. Legally, someone qualified must be accountable for what a worker is tested on — 'the AI wrote it' is not an acceptable answer. Technically, the model has no way to know if it is right. So the expert is the only correctness check in the whole system. When they approve, they re-enter their own password, and the system saves a fingerprint of the exact content approved. After that the question cannot be edited. And to be clear about the red bar — these are GxP-oriented controls, not a claim of compliance."

Words on this slide (see Part 3): SME · Electronic signature · Content hash · Immutability · Audit trail · 21 CFR Part

Slide 10 — Adaptive Learning (main contribution) (100 sec)

What is on the slide: A flow, the four priority bands, and the three governing numbers.

What to say:

"This is the core. For each section I take the learner's answers, newest first, and calculate an accuracy where recent answers count more than old ones. Why? Because a plain average cannot show change. Someone who got five wrong and then five right has clearly improved — but their average is still 50%, so the screen showed 'recent: 100%' next to a 'HIGH priority' warning and contradicted itself. With recency weighting they score 66.7% and move down a band. Then the rules: never assessed is checked first and always counts as HIGH, because no data must never be mistaken for competence. Below 60% is HIGH, below 80% is MEDIUM, above that is LOW, and three passes in a row retires the section."

Words on this slide (see Part 3): Recency weighting · Half-life · MIN_EVIDENCE · Mastery streak · Priority bands

Slide 11 — What Kind of System Is This? (80 sec)

What is on the slide: Four layers, then three statements on a dark bar.

What to say:

"I want to be precise, because it would be easy to overclaim. It is a hybrid, in four layers. Layer one, the decision, is rule-based — fixed thresholds that I chose, not learned. Layer two is the statistic those rules run on, which IS calculated from data. Layers three and four, Elo and FSRS, genuinely do estimate their parameters from learner responses as they arrive. So: the language model does not decide what gets re-trained. The adaptive engine is not a trained neural network. And it is adaptive because measured performance changes what content comes next."

Words on this slide (see Part 3): Hybrid · Rule-based · Statistical · Model-based · Online estimation

Slide 12 — The Complete Adaptive Loop (80 sec)

What is on the slide: A seven-step loop, with Elo and FSRS panels below, and a grey box.

What to say:

"That is the full loop, and it returns to the start. Elo and FSRS run alongside it, not inside it. And the grey box is my favourite detail: FSRS gives a 'how well do you remember this' score, and immediately after a test that score is 1.0 for everything — for a section you just failed AND one you just passed. It literally cannot tell them apart at the exact moment I need it to. That is why selection uses accuracy and FSRS only handles timing."

Words on this slide (see Part 3): Retrievability · $R(0,S) = 1.0$ · What vs When

Slide 13 — Results (70 sec)

What is on the slide: A bar chart 50 → 100, the demo numbers, and the verification results.

What to say:

"First the caveat: this is a verified controlled demonstration, not a user study. Now the result. The learner scored 33%. The system picked six questions instead of nine and excluded the section they already knew. After re-training, both weak sections went from 50% to 100% — fifty percentage points, calculated from their stored answers by comparing the older half of their history against the newer half. And on the engineering side: 221 tests passing, everything clean."

Words on this slide (see Part 3): Controlled demonstration · Learning gain · Regression tests

Slide 14 — Security, Validation and Testing (60 sec)

What is on the slide: Ten controls, the test count, and a red bar.

What to say:

"Do not read all ten. Pick three: grading happens on the server, never in the browser. The correct answer is never sent to the learner's browser at all — I checked the raw data, not just the screen. And a completed test cannot be submitted twice, so a training record cannot be overwritten. 221 tests. But note the red bar: tests prove the system behaves correctly; they do not prove regulatory compliance."

Words on this slide (see Part 3): Answer-key protection · Atomic claim / HTTP 409 · RBAC

Slide 15 — Limitations and Future Scope (80 sec)

What is on the slide: Five limitations in red, three columns of future work, and a priority bar.

What to say:

"I will give you the biggest limitation before you ask for it. The system works out which questions to offer, but it does not save that list. It checks that any submitted question belongs to the right SOP and is approved — but a modified app could send back a different approved question, or leave some out. So the adaptive decision is calculated and checked on the server, but not yet enforced. The fix is one extra database table, and it is first on my list. The ordering of my roadmap is deliberate: integrity first, then better measurement, then more intelligence."

Words on this slide (see Part 3): Offered-question binding · SOP versioning · IRT · BKT · Agentic AI

Slide 16 — Conclusion (50 sec)

What is on the slide: Seven points and four closing lines.

What to say:

"Go through the seven quickly, then slow right down for the last four lines: AI generates the content. Experts control the content. Learner data drives adaptation. And FSRS controls the timing. Thank you — I am happy to take questions."

Slide 17 — References (if asked)

What is on the slide: Seven academic references.

What to say:

"Only if asked. References three and six are the two I actually implemented — Elo and the memory model. Reference seven is the study that justifies compulsory expert review."

Part 3 — Every Technical Word, Explained

Grouped by topic rather than alphabetically, because the words make more sense together. Each entry gives a plain-English meaning, then the detail, then a sentence you can actually say.

3.1 The industry words

SOP (Standard Operating Procedure)

In plain words: A written instruction document telling a worker exactly how to do one task.

In a medicine factory everything is written down: how to enter a clean room, how to record a problem. Workers must be trained on the SOPs relevant to their job. In this project, an SOP is the input document.

Say it like this: *"An SOP is a controlled instruction document — every task in a regulated factory has one."*

GxP

In plain words: A family of 'good practice' rules that medicine manufacturers must follow.

The x stands for a category: GMP is Good Manufacturing Practice, GLP is laboratory practice, and so on. These rules require that staff are trained and that the training is recorded and provable.

Say it like this: *"GxP is the umbrella term for good-practice regulations in pharmaceutical manufacturing."*

GxP-oriented (NOT GxP compliant)

In plain words: It is built following the spirit of those rules, but nobody has certified it.

Real compliance requires a formal validation exercise, documented testing against a specification, and usually an audit. Your project has several of the technical controls but has not been through that process. This distinction is the single most important thing to get right verbally.

Say it like this: *"This system implements GxP-oriented controls. I am not claiming it is a validated or compliant system."*

21 CFR Part 11

In plain words: The specific American regulation covering electronic records and electronic signatures.

It says that if you replace paper signatures with electronic ones, the electronic signature must be reliably linked to the record and attributable to a specific person. Your approval screen mentions it. Always describe your implementation as 'Part 11-style', never as compliant.

Say it like this: *"We implement Part 11-style controls: password-verified approval bound to a content hash."*

SME (Subject Matter Expert)

In plain words: The qualified human who checks and approves each AI-written question.

In your system this is a specific user role. They can see the correct answers and the review queue; ordinary learners cannot. Nothing reaches a learner without SME approval.

Say it like this: *"The SME is the only correctness control in the pipeline — the AI drafts, the human decides."*

CAPA

In plain words: Corrective and Preventive Action — the formal process for handling something going wrong.

It is simply the name of one of the three sections in your demo SOP. You do not need to explain the process itself; just know it is a section title.

Say it like this: *"CAPA is one of the three sections in the demo procedure."*

3.2 The AI words

LLM (Large Language Model)

In plain words: The kind of AI that reads and writes text, like ChatGPT.

In your project the LLM's only job is to draft quiz questions from a passage of SOP text. It does not grade anyone, and it plays no part in deciding what a learner gets re-trained on.

Say it like this: *"The LLM generates assessment content. It has no vote in any adaptive decision."*

NVIDIA NIM

In plain words: The service that runs the AI model for you, over the internet.

Instead of running a large model on your own computer, you send the text to NVIDIA's service and get the generated questions back. The specific model used is Llama 3.1 8B — '8B' means eight billion parameters, i.e. a mid-sized model.

Say it like this: *"Generation runs on NVIDIA NIM, using Llama 3.1 8B."*

Chunking

In plain words: Cutting a long document into smaller pieces.

Your system tries three methods in order. First it looks for numbered headings, because real SOPs have them — this is the best result because each piece is a real section. If that fails it uses AI embeddings to find where the topic changes. If that also fails it just cuts every N characters. The method actually used is saved with each chunk.

Say it like this: *"A three-tier chunking cascade: heading-aware first, then semantic, then fixed-length."*

Embeddings / semantic chunking

In plain words: Turning text into numbers so a computer can measure whether two passages are about the same thing.

Used only as the fallback if heading detection fails. Passages that are about similar topics produce similar numbers, so the system can find the point where the subject changes and cut there.

Say it like this: *"Semantic chunking uses embeddings to detect where the topic shifts."*

Provenance

In plain words: Knowing exactly where something came from.

Every generated question stores a link to the specific chunk of SOP text it was written from. A reviewer can click and see that exact passage. This is what makes the whole adaptive system possible.

Say it like this: *"Every question stores a reference to its source passage, so provenance is always recoverable."*

Entailment (and why you do NOT have it)

In plain words: Automatically proving that the answer really does follow from the passage.

This is the honest gap. You can show WHERE a question came from, but nothing in the system mechanically verifies that the marked-correct answer is actually supported by that text. A human does that. Admitting this openly earns marks; being caught overclaiming loses them.

Say it like this: *"This is provenance plus a prompt constraint. It is not verified entailment — the SME is the check."*

Hallucination

In plain words: When an AI confidently writes something that is simply wrong.

This is the risk your SME gate exists to catch. Your cited study of 71 papers found error rates in AI-written questions from under 1% up to 45%.

Say it like this: *"I mitigate hallucination at three layers and prevent it at one — the human."*

Deduplication (lexical, not semantic)

In plain words: Removing near-identical questions before a reviewer sees them.

Your system compares the actual words used. If two questions share enough vocabulary they are treated as duplicates. It will NOT catch two questions that mean the same thing using completely different words — that is on the future list.

Say it like this: *"Duplicate detection is lexical. Semantic duplicate detection is future scope."*

Deterministic offline fallback

In plain words: A backup question generator that runs when the AI service is unavailable.

It does not invent anything: it lifts the correct answer word-for-word from the SOP text. Lower quality, but it cannot hallucinate. It tries the AI three times first.

Say it like this: *"After three retries it falls back to a deterministic generator that takes the answer verbatim from the SOP."*

3.3 The adaptive-learning words — the most important section

Adaptive learning

In plain words: The content a learner gets next changes based on how they performed.

Note what it does NOT mean: it does not mean a neural network, and it does not mean the AI decides. In your system it means measured per-section performance changes which questions are offered next.

Say it like this: *"It is adaptive because measured learner performance changes future content selection."*

Section-level mastery

In plain words: Tracking how well a learner knows each SECTION separately, not the document as a whole.

This is the core idea. Instead of one score for the SOP, there is a separate record per (learner, section). This is why the system can exclude one section and re-train two others.

Say it like this: *"The unit of adaptation is the SOP section, not the document."*

Recency weighting

In plain words: Recent answers count more than old answers.

The newest answer has full weight; older answers count progressively less. This replaced a plain lifetime average, and it fixed a real bug: a learner who went 0-out-of-5 then 5-out-of-5 has obviously improved, but their plain average is still 50%, so the screen said 'recent: 100%' next to a HIGH-priority warning — contradicting itself. Weighted, they score 66.7%.

Say it like this: *"Priority runs on a recency-weighted accuracy, because a flat average cannot represent change."*

Half-life = 5

In plain words: An answer five questions ago counts half as much as the newest one.

This is the speed of the recency weighting. Five was chosen because the interface already showed 'recent accuracy' over the last five answers, so 'recent' means the same thing everywhere in the system.

Say it like this: *"The half-life is five answers, matching the recent-accuracy window shown in the interface."*

MIN_EVIDENCE = 3

In plain words: You need at least three answers before the system will declare a section safe.

One lucky correct answer should not retire a whole section. Note the deliberate asymmetry: strong performance on fewer than three answers is capped at MEDIUM and cannot exclude the section — but WEAK performance on one or two answers still counts as HIGH. That is because over-training costs a few extra questions, while under-training could leave someone unqualified.

Say it like this: *"The evidence gate is asymmetric on purpose: under-training is the more costly error."*

Priority bands (HIGH / MEDIUM / LOW / NONE)

In plain words: The four verdicts the system gives each section.

Below 60% is HIGH — a real weakness. Between 60% and 80% is MEDIUM — needs reinforcement. At or above 80% is LOW — performing well. NONE means mastered and retired. HIGH and MEDIUM sections get selected for re-training; LOW and NONE do not.

Say it like this: *"Below 60% is HIGH, below 80% is MEDIUM, at or above 80% is LOW, and mastered sections are retired."*

Never assessed → HIGH

In plain words: A section nobody has been tested on is treated as a weakness, and checked first.

This fixed a real bug where sections that had never been tested were invisible to the system, so a learner could be re-trained forever on the one section they had been tested on.

Say it like this: *"Absence of evidence must never be read as competence."*

Mastery streak = 3

In plain words: Three passing assessments in a row retires a section from re-training.

This is what moves a section to priority NONE. One failure resets the streak to zero.

Say it like this: *"Three consecutive passing assessments mark a section mastered."*

Learning gain

In plain words: The measured improvement, in percentage points.

Calculated by splitting that section's answer history in half and comparing the older half against the newer half. It needs at least four answers, because below that there are not two halves worth comparing. In the demo it is 50% to 100%, so +50 points.

Say it like this: *"Gain compares the oldest half of the answer history against the newest half. It is measured, not asserted."*

3.4 The two model-based components**Elo rating**

In plain words: The chess ranking system, applied to learners and questions.

Both a learner and a question start at 1500. Every answer is treated as a match between them. Answer a hard question correctly and your rating rises a lot; the question's rating falls slightly. Two different speeds are used: the learner's rating moves fast (K=32) because we need a useful estimate from few answers, while a question's rating moves slowly (K=16) because it is shared across every learner.

Say it like this: *"Elo estimates learner ability and question difficulty online — the same family as IRT, with a cheaper update rule."*

K-factor

In plain words: How fast a rating moves after each answer.

Learner K=32, question K=16. That is the only difference, and the reason is above.

Say it like this: *"Learner K is 32, question K is 16 — the learner must adapt from few answers; a question is shared."*

FSRS

In plain words: The algorithm that decides WHEN a learner should review something again.

It stands for Free Spaced Repetition Scheduler. It models memory using three quantities — Difficulty, Stability and Retrievability — and schedules the next review for the moment your chance of remembering drops to about 90%. It replaced a fixed 1/2/4/7/14/30-day ladder, which treated all material identically.

Say it like this: *"FSRS models memory and schedules the next review at 90% predicted retention."*

Retrievability, and $R(0,S) = 1.0$

In plain words: Your probability of remembering something right now — and the reason FSRS cannot choose what to study.

This is the sharpest technical point in your project. Immediately after a test, zero time has passed, so retrievability is 1.0 for EVERY section — including one you just failed and one you just passed. It cannot tell them apart at exactly the moment you need it to. That is why accuracy chooses the content and FSRS only chooses the timing.

Say it like this: *"Retrievability is 1.0 for every stability at zero elapsed time, so FSRS mathematically cannot select content."*

Spaced repetition

In plain words: Reviewing material at increasing intervals so it sticks in long-term memory.

The general technique FSRS implements. Review something just before you would forget it, and the memory strengthens.

Say it like this: *"Spaced repetition schedules review just before predicted forgetting."*

WHAT vs WHEN

In plain words: Two different questions answered by two different parts of the system.

Adaptive selection answers WHAT should be trained (based on accuracy). FSRS answers WHEN it should be reviewed (based on memory). A section can be flagged HIGH priority but not yet due — so the interface shows both, and never promises a quiz it cannot deliver.

Say it like this: *"adaptive.py answers what; fsrs.py answers when. Both must be true for training to be available now."*

3.5 The classification words (asked in almost every viva)

Rule-based

In plain words: Decisions made by fixed if-then rules that a human wrote.

Your priority layer is this: below 60% is HIGH, and so on. Those numbers were chosen by you, not learned from data. This is a strength in a regulated setting, because a person can check the reasoning.

Say it like this: *"The decision layer is rule-based, and deliberately so — a threshold and a count can be audited."*

Statistical

In plain words: A number calculated from data.

Your recency-weighted accuracy is a statistic. It is computed from real learner answers, but it is a calculation, not a trained model.

Say it like this: *"The rules operate on a statistic estimated from the learner's actual responses."*

Model-based / online parameter estimation

In plain words: A model whose internal numbers update as new data arrives.

Elo and FSRS do this. Elo updates ability and difficulty after every answer; FSRS updates memory state. This IS learning from data — but it is not the same as training a model on a dataset beforehand.

Say it like this: *"Elo and FSRS estimate parameters online, updating from each response as it arrives."*

Hybrid adaptive system

In plain words: The correct name for your architecture.

It combines all three of the above: rule-based decisions, on a statistical signal, supported by two model-based components. Say this phrase and you have answered the classification question correctly.

Say it like this: *"It is a hybrid: a rule-based decision layer over a statistical signal, with two model-based components."*

Why it is NOT machine learning

In plain words: Nothing was trained on a dataset beforehand.

There is no training set, no loss function, no gradient descent, no trained model file. Claiming ML and then being unable to name your training data would badly damage your credibility. The honest answer is stronger.

Say it like this: *"It is not machine learning in the trained-model sense. What I have is online parameter estimation."*

BKT, IRT, DKT (things you did NOT use)

In plain words: Three established learner-modelling techniques you deliberately did not implement.

BKT is Bayesian Knowledge Tracing, IRT is Item Response Theory, DKT is Deep Knowledge Tracing (neural). All need far more data per learner than a factory training system produces. Saying you read them and chose not to use them, with a reason, is much stronger than not knowing them.

Say it like this: *"I read them and declined them on data scale and auditability — Elo is the sample-efficient approximation."*

3.6 The software and security words

Django / DRF / React / Vite

In plain words: The tools the application is built with.

Django is the Python framework running the server; DRF (Django REST Framework) is the part that provides the API. React is the JavaScript library building the screens, and Vite is the tool that packages it. React runs in the browser; Django runs on the server.

Say it like this: *"Django and DRF on the backend, React on the frontend."*

Redis / Celery

In plain words: Tools for running slow jobs outside the main request.

Used for the AI generation tasks. Be honest about one detail: your code waits for these to finish rather than truly running them in the background, so it gives process isolation rather than genuine asynchrony.

Say it like this: *"Celery gives process isolation here; the request still waits for the task."*

Server-side scoring

In plain words: The grading happens on the server, never in the learner's browser.

This matters because anything running in a browser can be modified by the user. If grading happened there, every score, mastery record and adaptive decision downstream would be untrustworthy.

Say it like this: *"Grading is entirely server-side — the client never decides anything."*

Answer-key protection

In plain words: The correct answer is never sent to the learner's browser.

Learners and reviewers get different versions of the data. The learner's version has the correct-answer flag and the explanation removed entirely. You verified this by inspecting the raw network response, not just the screen.

Say it like this: *"Learners receive a different serializer that omits the correct-option flag entirely."*

Electronic signature + content hash

In plain words: The expert re-types their password to approve, and the system saves a fingerprint of what they approved.

The fingerprint (a SHA-256 hash) is a short code calculated from the exact content. Change one character and the code changes completely — so tampering is detectable. This is why approved questions cannot be edited afterwards.

Say it like this: *"Approval re-verifies the reviewer's password and stores a SHA-256 hash of the exact approved content."*

Immutability

In plain words: Once approved, a question cannot be changed.

Attempts to edit or delete are refused with an error (HTTP 403). If content could change after signing, the signature would silently become meaningless.

Say it like this: *"Approved content is immutable — edit and delete both return 403."*

Audit trail

In plain words: A permanent log of who did what and when.

Your system records 17 kinds of action. Be honest about the limit: it is append-only by convention, not tamper-proof — there is no cryptographic chaining yet.

Say it like this: *"Seventeen attributed action types. It is not yet tamper-evident, and that is on the roadmap."*

RBAC (Role-Based Access Control)

In plain words: What you can see depends on your role.

Three roles: Admin, SME reviewer, and Learner. Menus and data are filtered by role, and enforced on the server rather than merely hidden in the interface.

Say it like this: *"Access is scoped by role, and enforced server-side rather than only hidden in the UI."*

Atomic claim / HTTP 409

In plain words: Stops the same test being submitted twice.

The system marks the attempt complete in one indivisible database operation, so if two submissions arrive at once only one can win. The loser gets a 409 Conflict error. This protects the training record from being overwritten.

Say it like this: *"An atomic compare-and-set claims the attempt, so a resubmission gets 409 rather than overwriting a record."*

Regression test

In plain words: A test that makes sure a bug you fixed does not come back.

You have 221 of these overall. Each bug you found has a test named after it. This is the strongest evidence that the project was built through verification rather than assumption.

Say it like this: *"Every defect I found has a regression test named for the bug it prevents."*

Offered-question binding (your biggest limitation)

In plain words: The system does not save the exact list of questions it offered a learner.

It checks that a submitted question belongs to the right SOP and is approved — but not that it was in the list originally offered. So a modified app could substitute a different approved question, or leave some out. The adaptive decision is therefore calculated and checked, but not enforced. The fix is one extra database table.

Say it like this: *"The adaptive decision is computed and validated server-side, but the offered set is not pinned yet."*

Part 4 — The Questions You Will Be Asked

Read the question, try to answer it out loud, then check. These are the twelve most likely.

Q1. Is your adaptive learning rule-based or machine learning?

"Both, and I will be precise. The decision layer is rule-based — fixed thresholds at 60% and 80%, a minimum-evidence gate of three, a mastery streak of three. Those I chose; they were not learned. But they are applied to a statistic calculated from data — a recency-weighted accuracy. And two components genuinely estimate parameters online from responses: Elo for ability and difficulty, and FSRS for memory. So it is a hybrid. It is adaptive; it is not machine learning, because nothing is trained offline — there is no dataset and no loss function."

Q2. Why didn't you use a neural network?

"Two reasons. Data scale: deep knowledge tracing needs thousands of interactions per learner, and a factory training system produces a few dozen — the model would fit noise and I would have no way to tell. And auditability: in a regulated setting, when the system says re-train this person on CAPA, that has to survive being questioned. 'Zero out of three, below the sixty percent threshold' survives. A neural network's output does not. With more data I would go Bayesian Knowledge Tracing, then IRT, then neural — interpretability first."

Q3. Then why is this an AI project at all?

"The generation half is genuinely AI — a language model drafting assessment items from controlled text, plus an embedding model for semantic chunking. And two adaptive components are parameter-estimating models; Elo is a latent-trait model, the same family as IRT. But the more interesting problem is how you put a probabilistic system inside a regulated workflow safely — where the human gate goes, and what has to stay deterministic and auditable. That is most of the work."

Q4. How is this different for two different learners?

"Two learners get verifiably different question sets, and there is a test named for exactly that. Everything is keyed on learner and section. One weak on CAPA gets the CAPA questions; one weak on documentation gets those instead. Their Elo ratings diverge from the first attempt, and their FSRS schedules diverge too — so both the content and the timing differ. And the amount differs: six questions instead of nine."

Q5. How do you know the learner actually improved?

"I split that section's stored answers in half and compare the older half against the newer half, needing at least four answers. In the demo, 50% to 100% — fifty points, on both weak sections. But I will be clear about the limit: that is a within-learner before-and-after measurement, not a controlled experiment. There is no control group, so I can show they improved on the material the system selected — I cannot show they would not have improved by re-reading the whole SOP. That needs an A/B trial."

Q6. How do you stop the AI generating wrong questions?

"I do not prevent it — I mitigate at three layers and prevent at one. Constrained generation, one section per prompt, no outside knowledge. Provenance, so the reviewer sees the source passage. And a deterministic fallback whose answer is lifted from the SOP text. The prevention layer is the SME: nothing reaches a learner without a qualified human approving it under signature. A review of 71 studies found error rates up to 45%, which is exactly why that gate is compulsory rather than optional."

Q7. Why is SME review necessary if you have AI?

"Two reasons. Regulatory: a qualified person must be accountable for what an operator is tested on — 'the model generated it' is not an answer an auditor accepts. Technical: the model has no ground truth. It produces plausible text conditioned on a passage; nothing in the pipeline can verify the answer is correct. So the SME is not bureaucracy on top of a working system — it is the only correctness control in it."

Q8. Why separate the adaptive engine from FSRS?

"Because they answer different questions, and they mathematically cannot merge. Selection answers what; FSRS answers when. The obvious move is to use FSRS's retrievability score for both — but at zero elapsed time it returns 1.0 for every stability value. Right after a test, a section you just failed and one you just passed both score 1.0. It cannot distinguish them at the exact moment I need it to. So accuracy selects and FSRS schedules."

Q9. Can a learner cheat the system?

"Partly, and I will be precise. The server checks that every submitted question belongs to this SOP, matches the role, is approved and is not duplicated — I can show the rejections. What it does not yet do is save the exact list it offered. So a modified client could substitute a different approved question, or leave some out. Bounded: same SOP, approved content, their own attempts, all audited. The fix is one join table. I did not ship a database migration on the most critical code path two days before review."

Q10. What is the biggest limitation?

"That the offered question set is not persisted. The adaptive decision is calculated and validated on the server, but it is not enforced. There is a second consequence too: because nothing records what was offered, a past attempt cannot be reconstructed exactly as the learner sat it, which weakens the training record. It is first on my roadmap."

Q11. Is this GxP compliant?

"No, and I would not claim it. It is GxP-oriented, with Part 11-style technical controls: password-verified approval bound to a content hash, an attributed audit trail, role-based access, immutable approved content. What is missing: tamper-evident audit storage, separation of duties, a qualification record, and any formal validation exercise."

Q12. Isn't this just a lot of if-else statements?

"The final classification is if-else, four branches, and I would defend that as correct for a compliance decision — it is auditable. But the interesting part is what is being branched on: a recency-weighted accuracy over an ordered answer history, per source section, gated on evidence sufficiency, reconciled against a spaced-repetition schedule, with two online-estimated rating systems feeding the mastery signal. The recency weighting alone fixed a real bug where the screen contradicted itself. The last step is if-else; everything that makes it correct happens before it."

If you are asked something you do not know

Say: *"I don't know that off the top of my head — let me tell you what I do know and where I would look."*

Never invent an implementation detail. Every document in this project is honest about its gaps, and being caught guessing costs far more than not knowing.